# OS Mid2 Practice Questions

1. Explain the Following:

| | | | |
|---|---|---|---|
| TLB | Stubs | Zero fill on demand | Socket with example |
| MMU | Hole | Belady's anomaly | compaction |
| Lazy swapper | Memory stall | Dirty bit | Access control list |
| Role of Index block in indexed allocation | List any 4 operations on a file | Limitation of single-level directory | Pupose of i-node in Unix file system |

2. Explain differences between following:

| | |
|---|---|
| Independent *vs* Cooperating process | Ordinary vs Named pipe |
| Blocking vs Non-blocking communication link | Absolute vs Relocatable address |
| Synchronous vs asynch communication link | Dynamic loading vs Dynamic linking |
| Logical vs Physical memory | Pages vs Frames |
| Continuous vs linked vs Indexed allocation | Internal vs External Fragmentation |
| Single-level vs multi-level vs tree-structured vs Acyclic graph structured directory | |

3. What is a Race condition? Explain with an Example.
4. Draw & Explain the Shared memory and Message passing models in Inter-process communication.
5. What are Remote procedural calls? Draw & explain an example.
6. Draw & explain concept of segmentation with an example? What is the purpose of segment table?
7. Draw & explain paging hardware with TLB.
8. Draw & explain the concept of shared pages in memory management.
9. Draw & explain the following structures of page table:
    a. Hierarchical
    b. Hashed
    c. Inverted
10. Write the steps in handling Page-fault along with the diagram.
11. Explain following page-replacement algorithms with an example:
    a. FIFO
    b. Optimal
    c. LRU
12. Draw & explain the concept of Copy-on-write in virtual memory management.
13. Explain various file attributes with an example
14. Describe the structure of a disk and explain how disk capacity is calculated.
15. Explain the implementation of the i-node in UNIX with a neat diagram.

16. Describe and compare different methods of free space management in file systems
17. Discuss the purpose and usage of read(), write(), lseek() and stat() with syntax and examples
18. Solve problems based on the following:
    a. Internal fragmentation
    b. Effective Access Time (EAT)
    c. Bits needed to identify a sector on a disk
    d. Maximum file size supported in a UNIX file system

Sample questions for problems

18)a) Calculate Internal fragmentation if page size is 2048 bytes and process size is 72,766 bytes

18)b) Calculate EAT if, TLB hit ratio is 80% and memory access time (for TLB hit) is 10nsec (So TLB miss requires two memory access)

18)c) How many bits are needed to identify a specific sector on a disk with 4 platters, 100 tracks/platter, and 50 sectors/track?

18)d) Calculate the maximum file size supported in a UNIX file system with:

- 12 direct blocks,
- 1 single indirect,
- 1 double indirect,
- 1 triple indirect block,
- Block size = 1KB, and each block address = 4 bytes